M. Abdullah SO1223

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DMIR

Assignment # 02

Q1: Mapping and Monitoring RICE Crop

#Preferred satellite imagery:

→ Sentinel-2:

→ 10-20m spatial resolution, ideal for agricultural monitoring.

→ ~~10-20m~~ 5-day revisit time.

→ Landsat 8/9:

→ 30m resolution

→ 16 dys revisit cycle.

→ Long historical data availablitity, useful for multi-years analysis.

Time season:

→ Rice crop is South Asia (e.g Pakistan-india)

→ Sowing begins in June-July, harvest around October - November.

→ Monitoring should start before sowing (to map bare soil) and continue through vegetative, reproductive and ripening stages.

Vegetation indices:

→ NDVI

→ Tracks crop greenness over time.

What about early growth time?, which one be more suitable

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→ EVI, uses NIR, Red, Blue
Better in dense vegetation and cloudy areas.
→ NDWI,
Helps monitor crop water stress.

#Why:
These indices help identify planting dates, growth stages and stress using temporal trend.

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Q2: Built-up area / Bare soil indices.

Common Indices:- which bands are used and how? (equation)

1: ~~NDBI~~ NDBI
highlights urban/built-up areas

2: BI
Enhances bare soil areas while suppressing vegetation surfaces.

3: IBI
- combines NDBI, SAVI and MNDWI
- offers better accuracy for identifying urban land

Band Used:
NIR, SWIR, Red and Blue.

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Enhanced vegetation using three Bands

1: Enhanced vegetation index (EVI):

→ Formula: $2.5 \times (NIR - RED) / (NIR + 6 \times Red - 7.5 \times Blue + 1)$

→ Bands: ~~NIR~~ Red Blue.

2: MCARI (Modified chlorophyll Absorption in Reflectance index)

→ Formula: $\{(R700 - R670) - 0.2 * (R700 - R550)\} * (R700 / R670)$

→ Focuses on chlorophyll concentration

→ Bands: Red-edge, Green, Red.

3: TCARI (Transformed chlorophyll Absorption in Reflectance Index)

/ Bands ?

→ Similar to MCARI, Adjusts for leaf structure effects.

equation

These indices provide better accuracy for vegetation health and stress under varying conditions.

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